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Extracted Text

Ortial ferer has not any mark to direct him: hence after- momers are obliged to enlighten the crois-hairs artifi, in order to render them vifible, and this with- out letting the luminous body interfere with the objefe which they intend to view , and, for doing this, two ways are propofel.

1. The object glafs of the telcfore may be obliquely enlightened by placing a could near usit in our oblique frition, to that its smoke or flame may not interfere with the object. But if the object-glafs should happen to be pretty deep in the tube, it cannot be difficultly enlightened by this means i said bedties, if the telefcope is above fix feet loop, there will be a considerable difficulty in throwing a sufficient quantity of high upon the crois hairs.

2. By fome an opening is made in the tide of the tube near the focus of the object-glafs, though which the crois hairs are illuminated by means of a candle. But this method alfo is attended with inconveniences for the oblorer is incommeded by the light being the near his eyes, and the hair harmless become liable to accidents through their expelled frustration. An error also attends this method i, which is, that, according to the position of the light illuminating these hairs, they will appear in different situations. For example, when the horizontal hair is enlightened always be perceive a luminos line which may be taken for the hair itself, and which appears at its upper superficies. On the other hand, when the hair is enlightened underneath, the luminos line will appear as the under the face of i, and the corners will be the diameter of the hair, which often amounts to more than fix records. M. de la Hire, however, found a remedy for this convenience. He often obterred, that in monodhine nights, when the weather was a little foggy, the crois hairs were diflically fees; but when the heavens were ferene, they could occur by her corrected. Here therefore covered with a piece of gauge, or fine filk-craps, that end of the tube next the object-glafs i and this method succeeded to find the task at a good dilution from the telefope to enlightened the crapes, that the crats-hairs diflically appeared, and the fight of the fans was by no means obforced.

In making the oblorations, a model glafs mulb uefd for preserving the eyes i and which may be thus prepared. Take two equal and well political round prices of glafs into the farles of one of which, all round its limb, glue a palkboard ring: then put the other piece of glafs into the smoke of a lamp, taking it several times time, and putting it in again, left the heat thould break it, until the smoke be fo thick, that the lamp can favorably be free through it, but the smoke mult not be all over of the same thickets, that fito a place may be obtain anfering to the splendour of the fun. This being done, the glafs, thus blackened, mult e glued to the pathboard of abovementand, with its black fide next to the other glafs, that the smoke may not be ribbed off.

There are two kinds of observation relating to the flares, one is, when they are in the meridian; the other, when they are in vertical circles. If the polition of the meridian ke known, the quadrant wohl which the oblorates is much, mult he placed in the plane of that circle, and then the meridian altitudes are really obferred by means of the plumb-line. The meridian

altitude of a flat may likewise be had by a pendulum clock, if the exact time of the flats pulling by the meridian be known. It mult be obffered, however, therehas lines the fans of latitude a minute before and after their puffling the meridians, if they be not in on near the zenith i, but if they be, their altitudes mult be takes every minute when they are in or near the meridian, and then their greatest or their altitudes will be those in question. The polition of a given vertical circle mult he found by the following method. 1. The quadratus and telescope remaining in the frame situation wherein it was when the altitude of a flux, together with the time of its puffage by the interjection of the crois hairs in the focus of the object-glafs, was taken, we observe the time when the fan, or frame facet flar, whole longitude and latitude is known, arrives to the vertical bar in the telefops and from those the potion of the first vertical circle will be hard, and alfo the obferred flar's true place. But if the fun, or fome olertas, does not ask by the mouth of the tubes, and if a meridian liebe otherwise will drawn upon allows, or very level ground, in the place of obloration i a plan-like mull be fuufpared from a fixed place, at about 18 or 30 feet from the quadrata y under which a matmulb te mode on the floor, in a right line with the plumb-line. You mult next put a thin piece of trade of palefoard very near the object-glafs, in the middle of which there is a fits vertically placed, and puffing through the centre of the circular figure of the object-glafs. Now, by means of this fills, the before-mentioned plumb-line area to prevent, through the telefcope, which before could not be feny, because of its moments. Then the plumb-line mail be removed and fulpated, to that it be prevented in a right line with the vertical hair in the focus of the object-glafs, and a point marked on the floor directly under it. And if a right-line the drawn through this points, and that marked under the plumb-line before it was removal, the full line will meet the merididan draw upon the floor i, and do we shall have the polition of the vertical circle in which the observed flar's, with respect to the meridian, the angle where' may be measured in afleming known lengths upon the two lines from the point of concavity, for fi, through the extremities of thefe known lengths, a line or bafe be drawn, we shall have a triangle, which three tides being known, the angle at the verce may be found, which will be the angle made by the vertical circle and meridian.

Under the parallel Qaz waves, is flew the proper method of fixing that obfluent exactly in the meridian, but where the oblorer has no convivency of this kind, it will be proper to use a guarda quadrant; by means of which the altitude of a flare mulb be ob-hered a little before its puffage over the meridian, every minute, if polibbs, until its greatest or lealt altitude be bad, by which means, though we have not the true polition of the meridian, yet we know the meridian altitude of the flar. But although this method is very good, yet if a flatter splits by the meridian near the zenith, we cannot have its meridian altitude by repeated observations every minute, unless by chance, because in every minutes can now, the altriche arguments (z minutes of a degree i) and in thefe kind of observations, the inconvenient limitation of the oblerver, the variation of the flar's azimuth feveral de-

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Optical Instruments

ſerver hath not any mark to direct him: hence aſtronomers are obliged to enlighten the croſs-hairs artificially, in order to render them viſible, and this without letting the luminous body interfere with the object which they intend to view; and, for doing this, two ways are propoſed.

1. The object-glaſs of the teleſcope may be obliquely enlightened by placing a candle near to it in an oblique ſituation, ſo that its ſmoke or flame may not interfere with the object. But if the object-glaſs ſhould happen to be pretty deep in the tube, it cannot be ſufficiently enlightened by this means; and beſides, if the teleſcope is above ſix feet long, there will be a conſiderable difficulty in throwing a ſufficient quantity of light upon the croſs hairs.

2. By ſome an opening is made in the ſide of the tube near the focus of the object-glaſs, through which the croſs hairs are illuminated by means of a candle. But this method alſo is attended with inconveniencies; for the obſerver is incommoded by the light being ſo near his eyes, and the hairs themſelves become liable to accidents through their expoſed ſituation. An error alſo attends this method; which is, that, according to the poſition of the light illuminating theſe hairs, they will appear in different ſituations. For example, when the horizontal hair is enlightened above, we perceive a luminous line which may be taken for the hair itſelf, and which appears at its upper ſuperficies. On the other hand, when the hair is enlightened underneath, the luminous line will appear at the under ſurface of it; and the error will be the diameter of the hair, which often amounts to more than ſix ſeconds. M. de la Hire, however, found a remedy for this inconvenience. He often obſerved, that in moon-ſhine nights, when the weather was a little foggy, the croſs hairs were diſtinctly ſeen; but when the heavens were ſerene, they could ſcarcely be perceived: he therefore covered with a piece of gauze, or fine ſilk-crape, that end of the tube next the object-glaſs; and this method ſucceeded ſo well, that a link placed at a good diſtance from the teleſcope ſo enlightened the crape, that the croſs-hairs diſtinctly appeared, and the ſight of the ſtars was by no means obſcured.

In making ſolar obſervations, a ſmoked glaſs muſt be uſed for preſerving the eyes; and which may be thus prepared. Take two equal and well poliſhed round pieces of flat glaſs; upon the ſurface of one of which, all round its limb, glue a paſteboard ring: then put the other piece of glaſs into the ſmoke of a lamp, taking it ſeveral times out, and putting it in again, leſt the heat ſhould break it, until the ſmoke be ſo thick, that the lamp can ſcarcely be ſeen through it: but the ſmoke muſt not be all over of the ſame thickneſs, that ſo a place may be choſen anſwering to the ſplendour of the ſun. This being done, the glaſs, thus blackened, muſt be glued to the paſteboard ring abovementioned, with its black ſide next to the other glaſs, that the ſmoke may not be rubbed off.

There are two kinds of obſervations relating to the ſtars: one is, when they are in the meridian; the other, when they are in vertical circles. If the poſition of the meridian be known, the quadrant with which the obſervation is made, muſt be placed in the plane of that circle, and then the meridian altitudes are eaſily obſerved by means of the plumb-line. The meridian altitude of a ſtar may likewiſe be had by a pendulum clock, if the exact time of the ſtar's paſſing by the meridian be known. It muſt be obſerved, however, that ſtars have the ſame altitude a minute before and after their paſſing the meridian, if they be not in or near the zenith; but if they be, their altitudes muſt be taken every minute when they are in or near the meridian, and then their greateſt or leaſt altitudes will be thoſe in queſtion. The poſition of a given vertical circle muſt be found by the following method. 1. The quadrant and teleſcope remaining in the ſame ſituation wherein it was when the altitude of a ſtar, together with the time of its paſſage by the interſection of the croſs hairs in the focus of the object-glaſs, was taken, we obſerve the time when the ſun, or ſome fixed ſtar, whoſe longitude and latitude is known, arrives to the vertical hair in the teleſcope; and from thence the poſition of the ſaid vertical circle will be had, and alſo the obſerved ſtar's true place. But if the ſun, or ſome other ſtar, does not paſs by the mouth of the tube; and if a meridian line be otherwiſe well drawn upon a floor, or very level ground, in the place of obſervation; a plumb-line muſt be ſuſpended from a fixed place, at about 18 or 20 feet from the quadrant; under which a mark muſt be made on the floor, in a right line with the plumb-line. You muſt next put a thin piece of braſs or paſteboard very near the object-glaſs, in the middle of which there is a ſlit vertically placed, and paſſing through the centre of the circular figure of the object-glaſs. Now, by means of this ſlit, the beforementioned plumb-line may be perceived through the teleſcope, which before could not be ſeen, becauſe of its nearneſs. Then the plumb-line muſt be removed and ſuſpended, ſo that it be perceived in a right line with the vertical hair in the focus of the object-glaſs, and a point marked on the floor directly under it. And if a right-line be drawn through this point, and that marked under the plumb-line before it was removed, the ſaid line will meet the meridian drawn upon the floor; and ſo we ſhall have the poſition of the vertical circle in which the obſerved ſtar is, with reſpect to the meridian, the angle whereof may be meaſured in aſſuming known lengths upon the two lines from the point of concourſe; for if, through the extremities of theſe known lengths, a line or baſe be drawn, we ſhall have a triangle, whoſe three ſides being known, the angle at the vertex may be found, which will be the angle made by the vertical circle and meridian.

Optical Inſtruments

Under the article Quadrant, is ſhewn the proper method of fixing that inſtrument exactly in the meridian: but where the obſerver has no conveniency of this kind, it will be proper to uſe a portable quadrant; by means of which the altitude of a ſtar muſt be obſerved a little before its paſſage over the meridian, every minute, if poſſible, until its greateſt or leaſt altitude be had; by which means, though we have not the true poſition of the meridian, yet we know the meridian altitude of the ſtar. But although this method is very good, yet if a ſtar paſſes by the meridian near the zenith, we cannot have its meridian altitude by repeated obſervations every minute, unleſs by chance; becauſe in every minute of an hour, the altitude augments 15 minutes of a degree; and in theſe kind of obſervations, the inconvenient ſituation of the obſerver, the variation of the ſtar's azimuth ſeveral degrees

Figure 1: *
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